INJECTIVITY OF A MAP OF FINITE TYPE MODULES OVER A LOCAL BASE, TESTED ON THE ASSOCIATED POINTS OF THE FIBRE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Vanishing of a module with no weakly associated points, and weakly associated points of a submodule. Divisors, labels divisors-lemma-weakly-ass-zero (proof line 5566) and divisors-lemma-ses-weakly-ass (proof line 5569). Reduces injectivity to the vanishing of the weakly associated points of the kernel and places those inside the weakly associated points of the module.
- Weakly associated points of a flat finite type module, and specialization of associated points. Same chapter flat.tex, labels lemma-bourbaki-finite-type-general-base (proof line 5571) and lemma-associated-point-specializes (proof line 5576). Shows every weakly associated point of the module is a relative associated point, and that purity makes each of those the specialization of a point of the fibre, so injectivity there propagates.

Source reference. Original lemma 05AN: sources/stacks-chow/tags.txt, source lines 6856–6856. Navigation pointer for \ref{divisors-lemma-ses-weakly-ass} used inside the counted span (line 5569); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AP: sources/stacks-chow/tags.txt, source lines 6857–6857. Navigation pointer for \ref{divisors-lemma-weakly-ass-zero} used inside the counted span (line 5566); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L3: sources/stacks-flat/pinned-original-source.tex, source lines 5175–5175. Navigation pointer for \ref{lemma-associated-point-specializes} used inside the counted span (line 5576); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 05IJ: sources/stacks-flat/pinned-original-source.tex, source lines 4079–4079. Navigation pointer for \ref{lemma-bourbaki-finite-type-general-base} used inside the counted span (line 5571); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of finite type. Let $\mathcal{F}$ be a quasi-coherent sheaf of finite type on X. Assume S is local with closed point s. Assume $\mathcal{F}$ is pure along X_s and that $\mathcal{F}$ is flat over S. Let $\varphi : \mathcal{F} \to \mathcal{G}$ of quasi-coherent $\mathcal{O}_X$-modules. Then the following are equivalent*

(1) *the map on stalks φ_x is injective for all $x \in Ass_{X_s}(\mathcal{F}_s)$, and*
(2) *φ is injective.*

Proof. Let $\mathcal{K} = \mathrm{Ker}(\varphi)$. Our goal is to prove that $\mathcal{K} = 0$. In order to do this it suffices to prove that $\mathrm{WeakAss}_X(\mathcal{K}) = \emptyset$, see Divisors, Lemma 05AP. We have $\mathrm{WeakAss}_X(\mathcal{K}) \subset \mathrm{WeakAss}_X(\mathcal{F})$, see Divisors, Lemma 05AN. As $\mathcal{F}$ is flat we see from Lemma 05IJ that $\mathrm{WeakAss}_X(\mathcal{F}) \subset \mathrm{Ass}_{X/S}(\mathcal{F})$. By purity any point x' of $\mathrm{Ass}_{X/S}(\mathcal{F})$ is a generalization of a point of X_s, and hence is the specialization of a point $x \in \mathrm{Ass}_{X_s}(\mathcal{F}_s)$, by Lemma 05L3. Hence the injectivity of φ_x implies the injectivity of $\varphi_{x'}$, whence $\mathcal{K}_{x'} = 0$. □